# Designing and Exporting Precision-Fit 3D Protective Covers for Class A Surfaces for Mercedes-Benz U.S. International Process Engineers Without Advanced CAD Experience

Welcome! These instructions will teach you how to create custom protective covers for Mercedes-Benz (or any other company's) vehicle parts. In our environment, "Class A" refers to any surface the customer can see (like the exterior paint or interior leather) (Tanrıverdi, 2024). Scratches on these surfaces cost the company thousands in rework. By following this guide, you can design a custom-fit, 3D printed plastic guard that snaps onto these surfaces to protect them during tool use or assembly.

<u>Warnings:</u>

- Data Privacy: All CAD files imported from the Mercedes-Benz NX database are proprietary. Do not export these files to personal cloud storage or share them with outside vendors without an NDA.
- Clearance: Do not design the cover to be a "Zero-Fit." Plastic shrinks when 3D printed. Include around 0.3 mm offset described below to avoid poor fit with the surface.
- Common Errors: Many times with Class A parts, there are a lot of small surfaces and if you don't select one it can mess up the entire process. To avoid this, you may click and drag to draw a square to select a group of faces at once (see *Image 1* below).

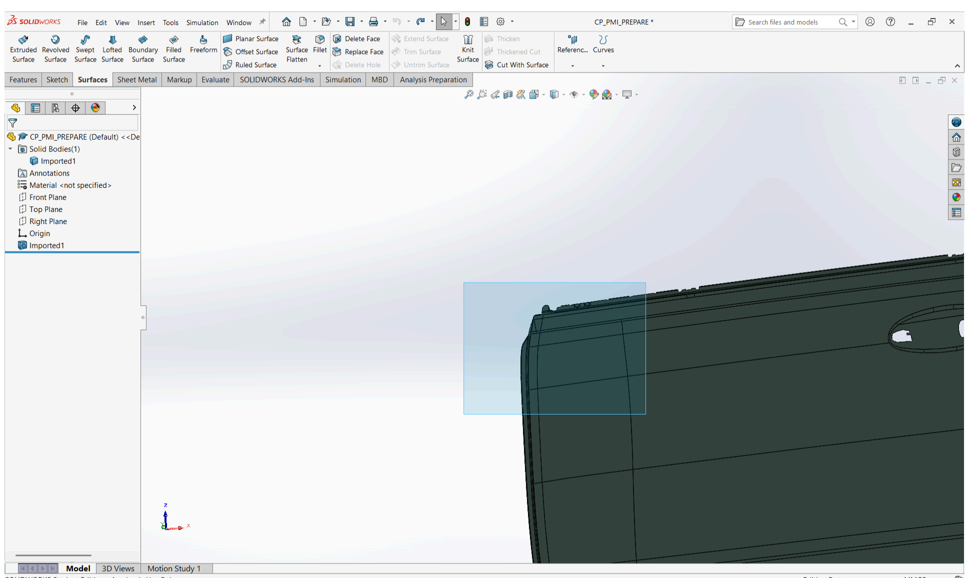


*Image 1: Selection of an Area in SOLIDWORKS*

<u>Necessary Background:</u>

- User should be able to complete simple functions in SOLIDWORKS
    - 2D Sketches (lines, shapes, planes…)
    - Smart Dimensions

    - Extruded Boss/Base
    - Extruded Cut
    - Fillet/Chamfer
- User should understand basic SOLIDWORKS navigation
    - Zoom
    - Pan
    - Rotate
    - Feature Tree

<u>Materials:</u>
Required:
- Computer capable of running SOLIDWORKS
- Computer mouse
- SOLIDWORKS software 2017 or later (any level of subscription is acceptable)
- CAD file of a Class A car part imported to SOLIDWORKS (there is already a set of technical instructions on how to get parts from NX to SOLIDWORKS)
    - *Note:* This file can be .step, .sldprt, or .x_t. MBUSI files will likely be .x_t

<u>Procedure:</u>

## Task 1: Import Model

### Step 1: Open the Reference Part

Open the model of the car part that is frequently being scratched. It should be exported from NX.

- If there are any questions on how to export a part from NX, see the other set of instructions located on the Mercedes-Benz Social Intranet

In SOLIDWORKS, locate the “File” tab in the top menu, click “Open”. See *Image 2* for reference.

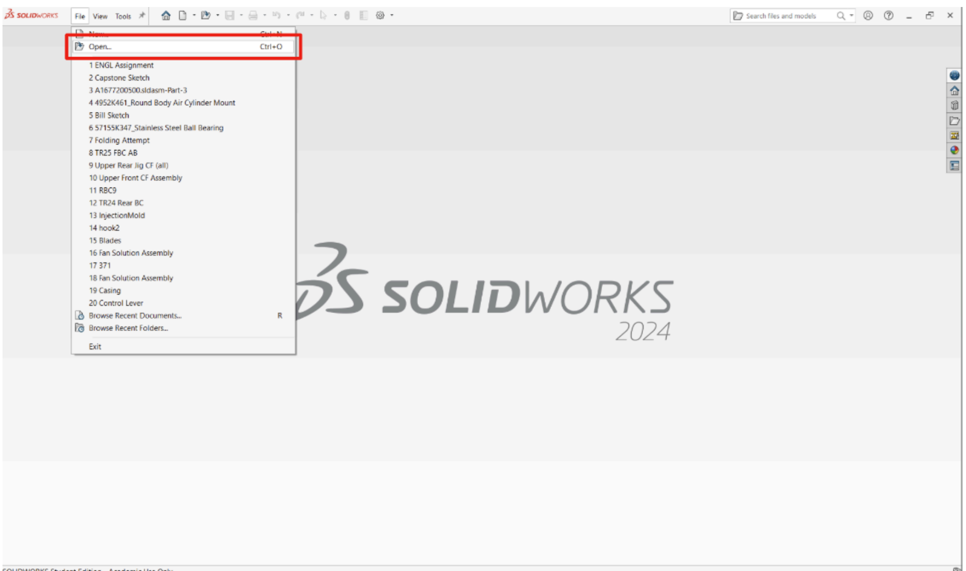


*Image 2: File → Open in SOLIDWORKS*

**Step 2: Isolate the Target Area**

Rotate the part to find the area that is commonly being scratched and needs a cover. A simple way to do this is by pressing the space bar to select certain faces. This is shown below in *Image 3*.

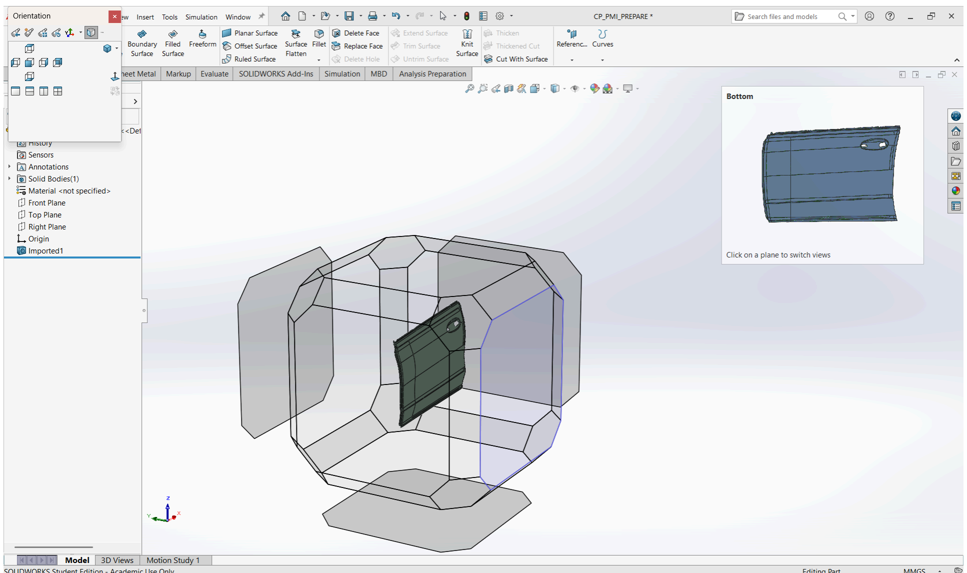


*Image 3: Cube to Select Certain Faces in SOLIDWORKS*

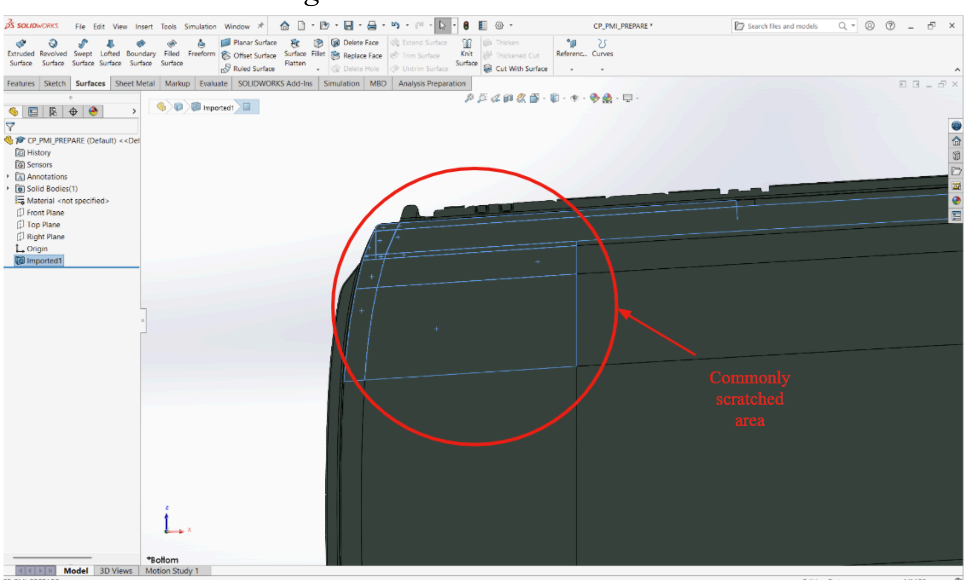


*Image 4: Highlighting the Commonly Scratched Area*

## Task 2: Create Base Geometry

**Step 3a: Insert "Surfaces" Tab into Toolbar**

Right click on the toolbar if you do not already see the “Surfaces” tab on the toolbar. Find “Surfaces” in the dropdown menu and click on it to add it. See *Image 5* for reference.

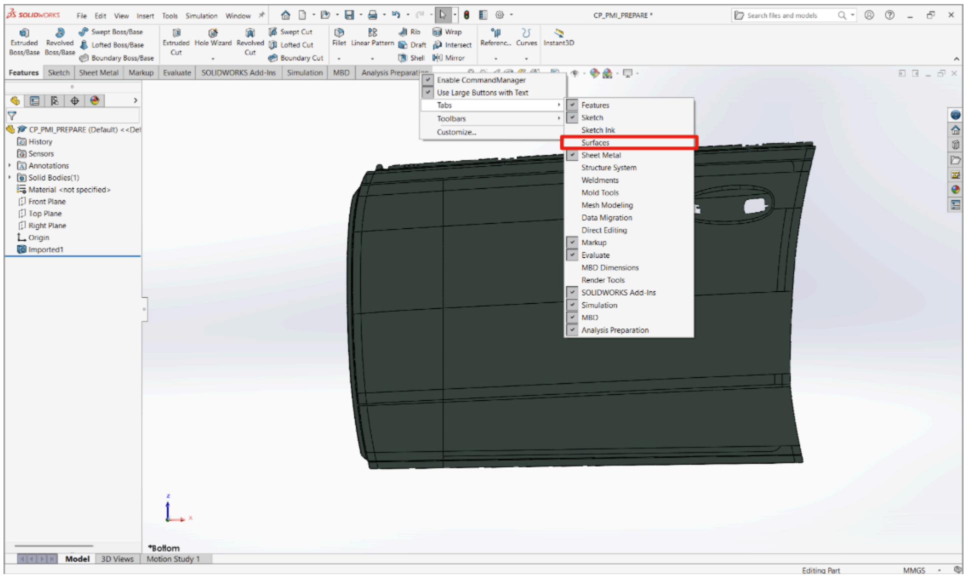


*Image 5: Adding the Surfaces Tab*

**Step 3b: Open “Offset Surface” Tool**

In “Surfaces”, locate and click on the “Offset Surface” feature (Dassault Systèmes, 2021). See *Image 6* for reference.

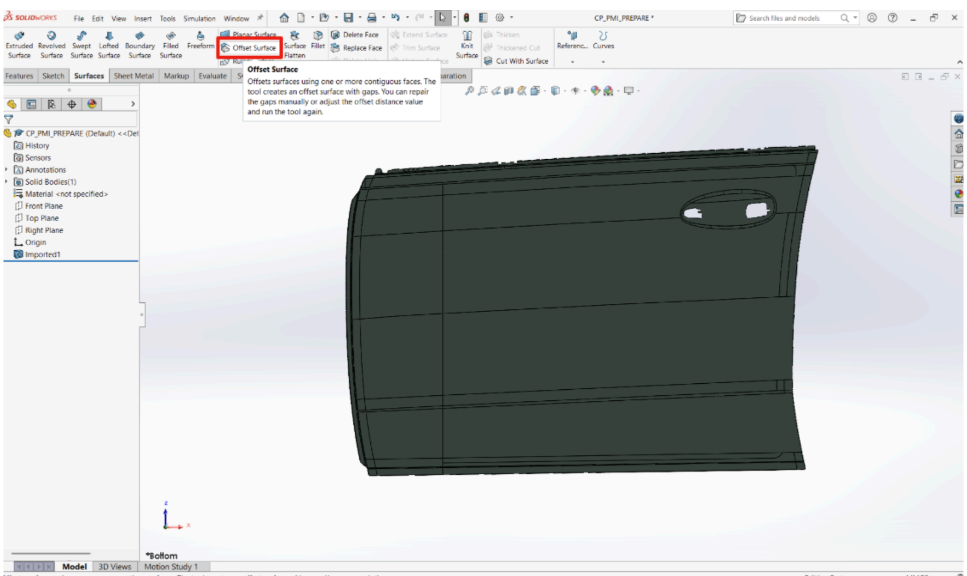


*Image 6: Offset Surface Feature*

**Step 4: Select Faces to Protect**

Click on the surfaces that commonly get scratched. You may either do this individually or as a group by highlighting a whole area as shown back in *Image 1*. See *Image 7* for selecting the faces individually.

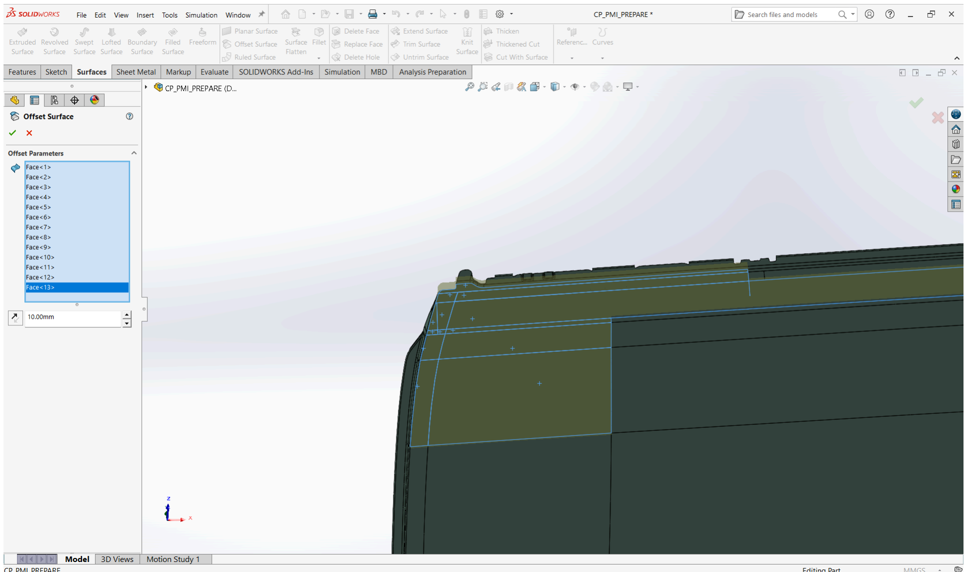

*Image 7: Selecting Faces to Protect*

**Step 5: Set the Clearance Gap**

Set the gap between the car part and the surface you are creating. Typically this should be somewhere from 0.2 to 0.5 mm (Xometry, 2023). I will use 0.3 mm for this example, but that is arbitrary. See *Image 8* to find where to insert the offset amount.

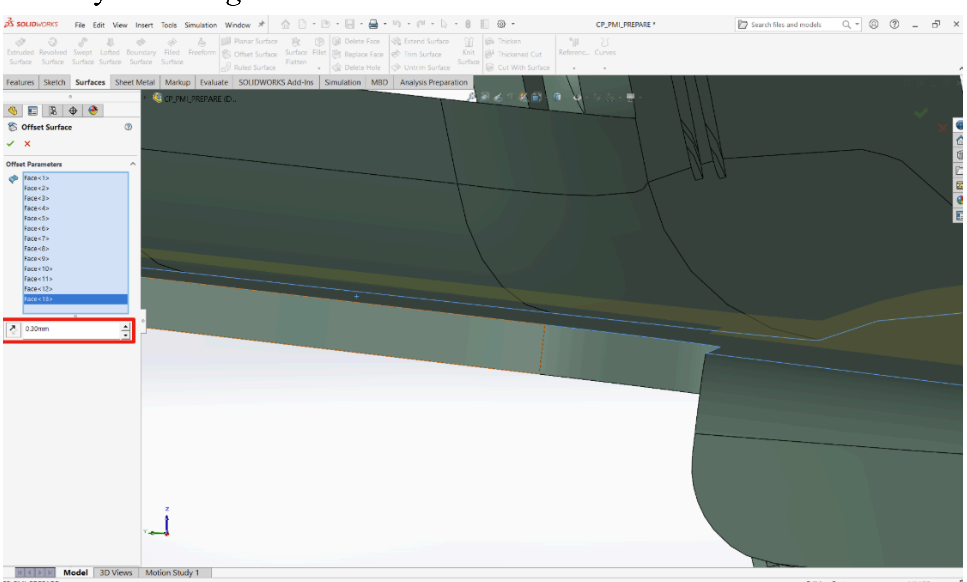

*Image 8: Offset Distance*

## Task 3: Thicken Geometry to Create Protective Cover

### Step 6: Open "Thicken Surface" Tool

Locate the "Thicken Surface" tool in the "Surfaces" tab and select it. See *Image 9* for reference.

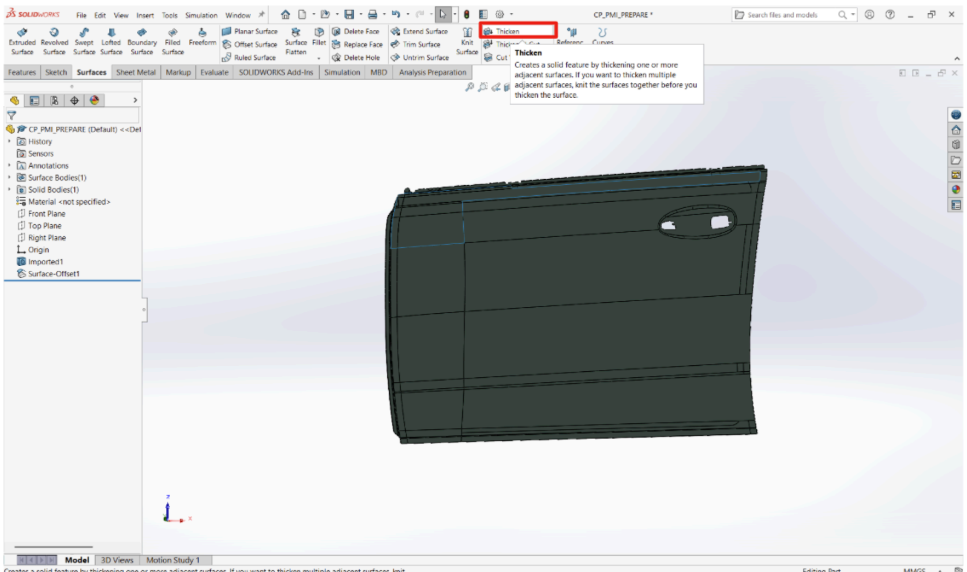


*Image 9: Thicken Feature in Surfaces Tab*

### Step 7: Set Thickness and Direction

Click on the surface you just created. One way to do this if you are struggling is to open the feature tree and select the Surface Offset feature that you just created. See *Image 10* for reference.

Set the thickness (3 mm is a good starting point unless that area of the car is often scratched by something that can slice through 3mm of plastic).

<u>Be sure to uncheck the merge box</u>, as this part needs to be different from the original car part. You will understand why in a later step. This is shown in *Image 11*.

Make sure the direction of the thickness extrudes out away from the original part, not in towards the part. This is also shown in *Image 11*.

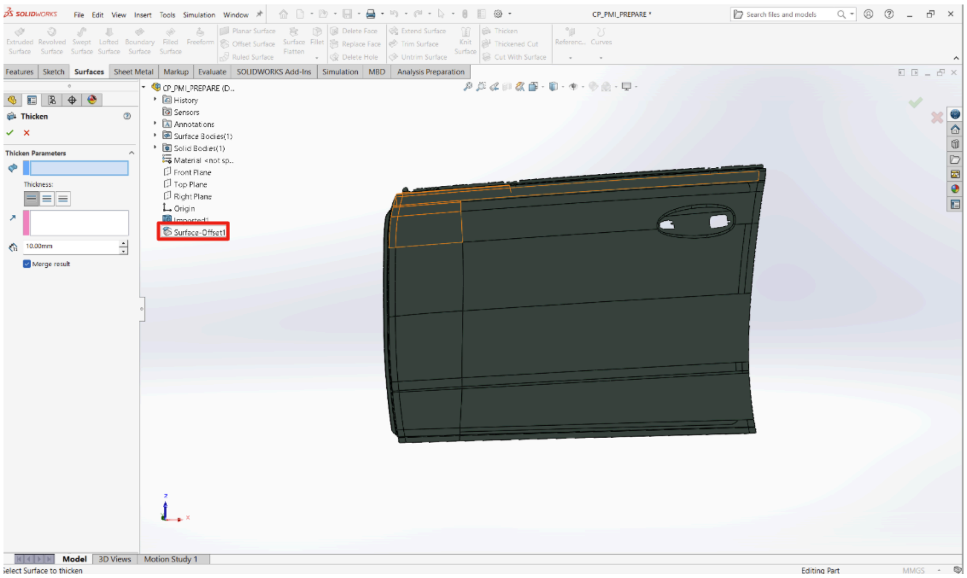

*Image 10: Selecting Surface Offset to Thicken*

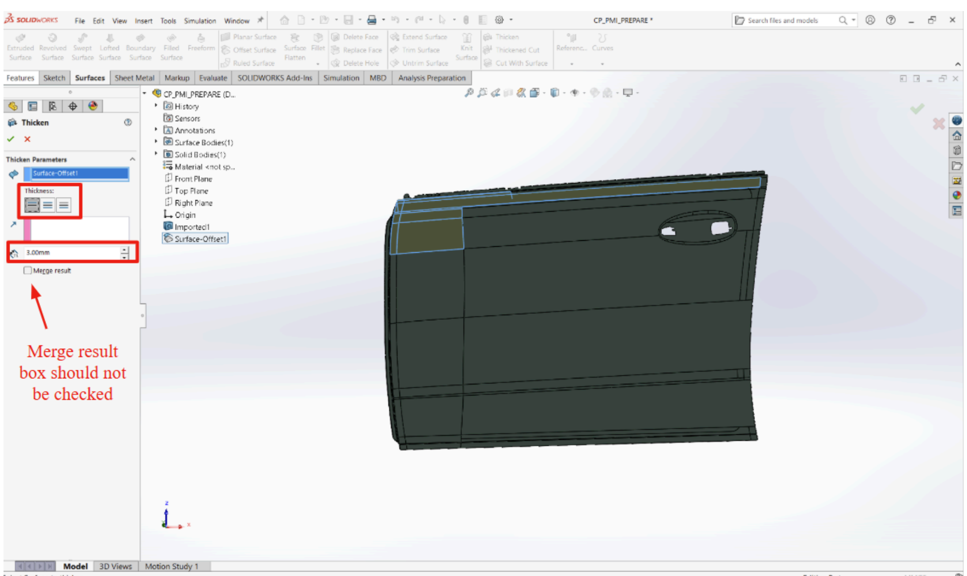


*Image 11: Thickness Direction and Unchecking "Merge result" Box*

## Task 4: Remove Initial Model

### Step 8: Identify Bodies (Should be at least two)

Look at the feature tree and find the section that shows bodies. Click to see the dropdown menu. See *Image 12* for reference.

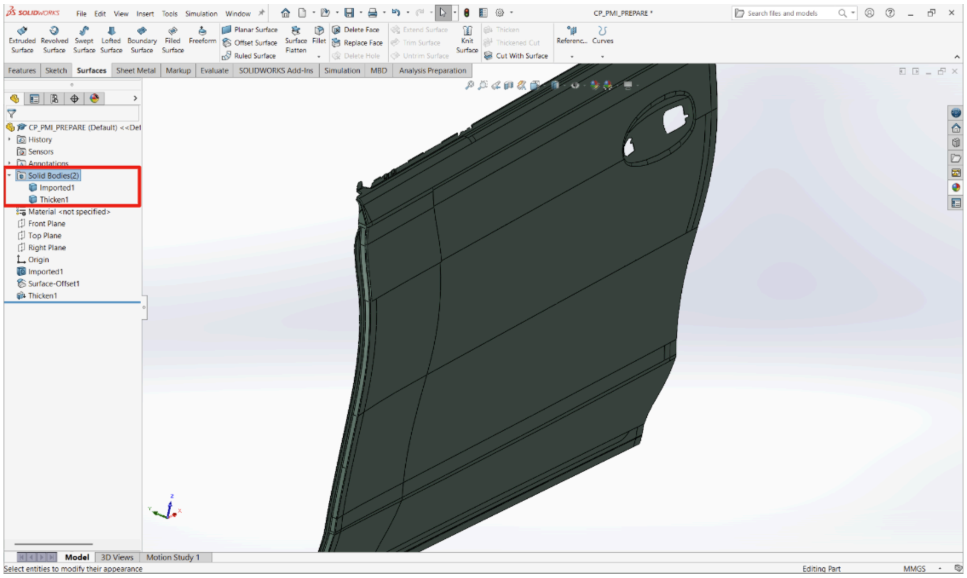

*Image 12: Select Bodies to Determine Which to Keep*

**Step 9: Delete Reference Part**

Right click the initial car part (in this case named "Imported 1") and click "Delete Body" so that you only have the new protective cover. This step is shown in *Image 13, Image 14* and *Image 15*.

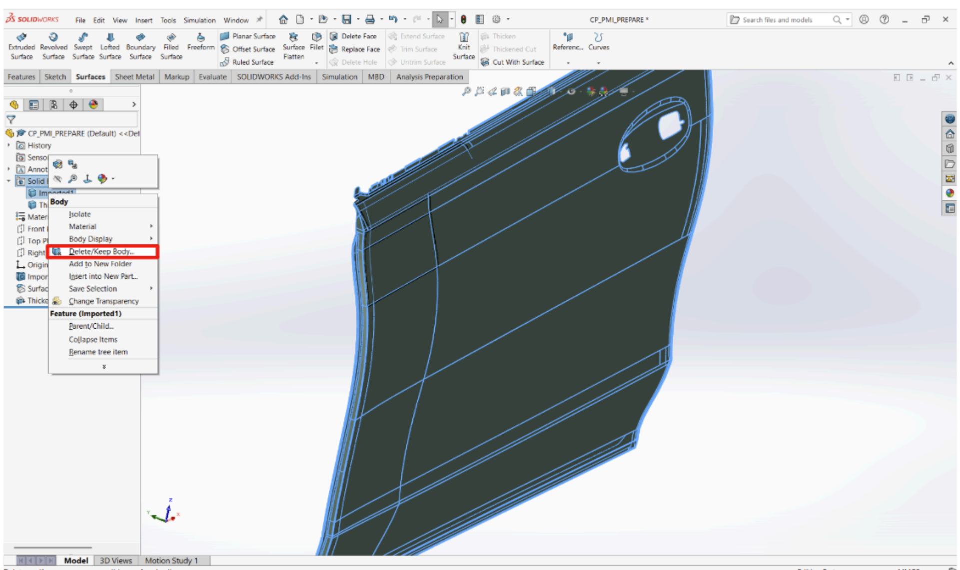

*Image 13: Delete/Keep Body Feature*

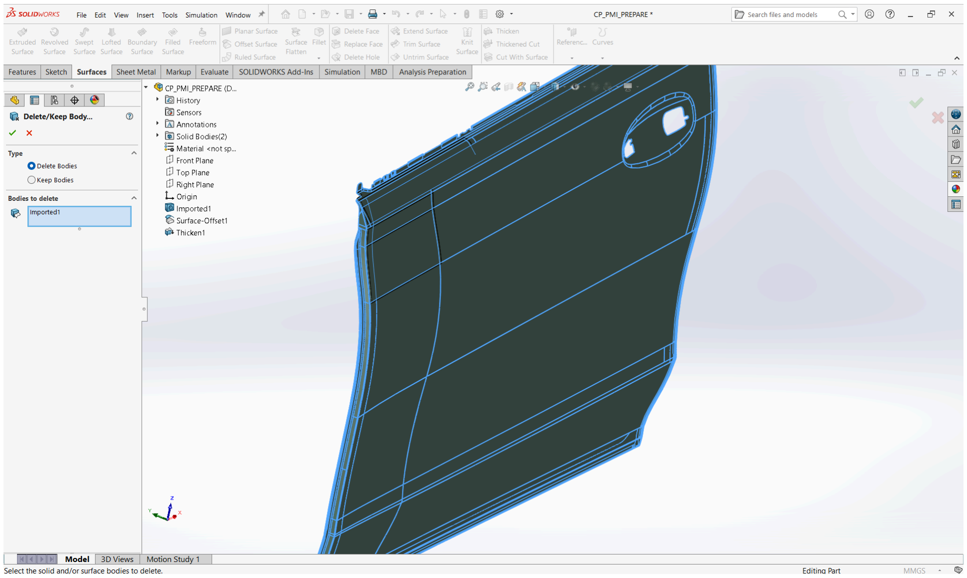

*Image 14: Selecting Delete Body and Selecting the Original Model*

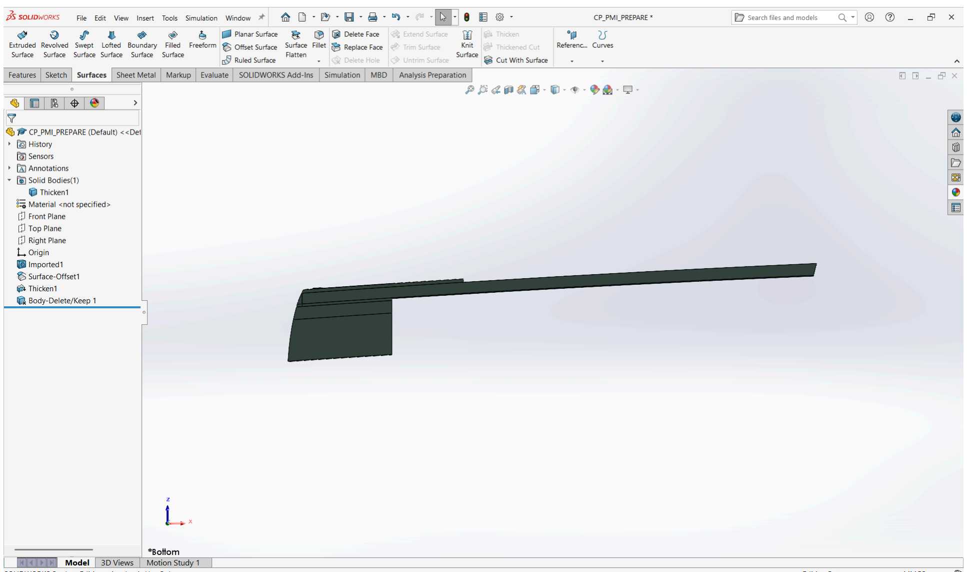

*Image 15: Final Part from Deleting the Original Model*

## Task 5: Clean Up and Export New Part for 3D Printing

### Step 10: Trim to Size

Trim any excess material from extended surfaces (Dassault Systèmes, 2026). Ensure that the cover is not too large in any direction for what you need. This should be something you know, but I include *Image 16, Image 17,* and *Image 18* just in case.

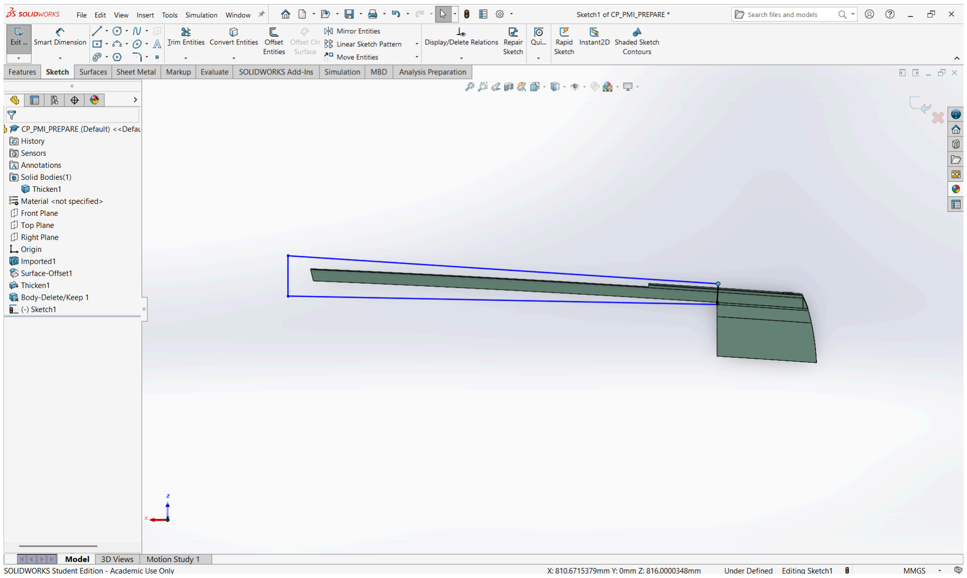

*Image 16: Sketch to Cut Excess Material*

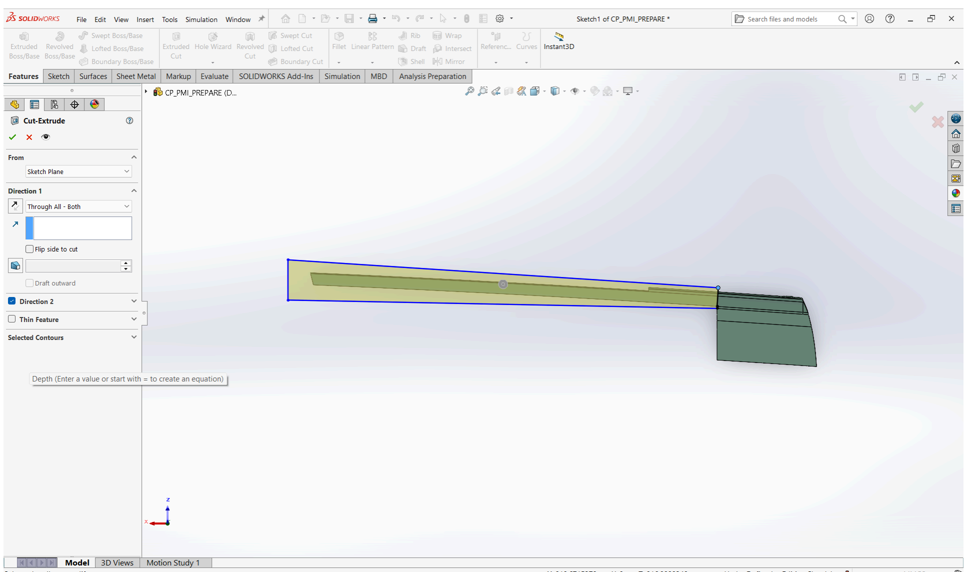

*Image 17: Extruded Cut Feature to Remove Excess Material*

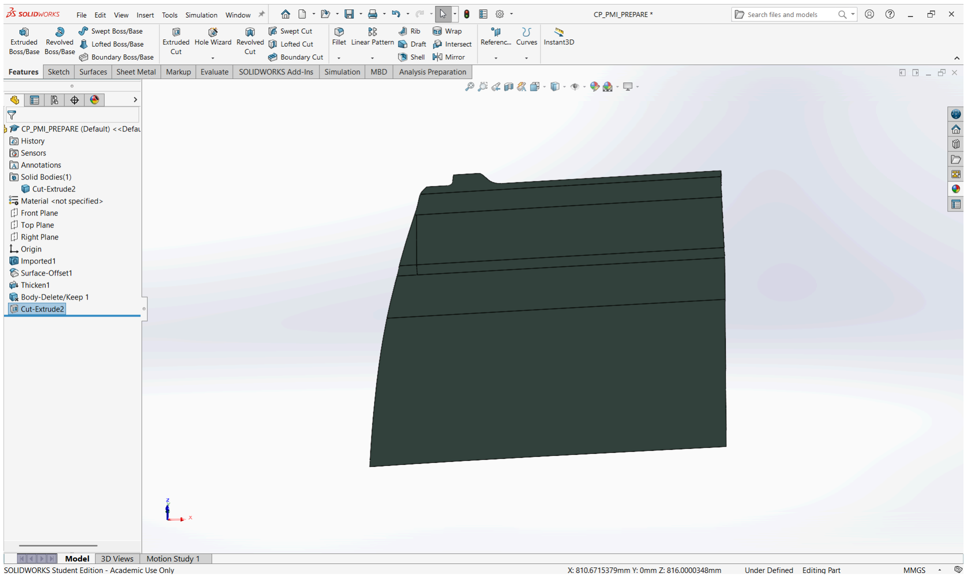


*Image 18: General Final Shape of Protective Cover*

## Step 11: Apply Fillets

Remove most if not all sharp corners to help with strength and/or printability. Part of this is also just to improve the physical appearance, making the part look more professional. See *Image 19* for reference. The dimensions are arbitrary for this example.

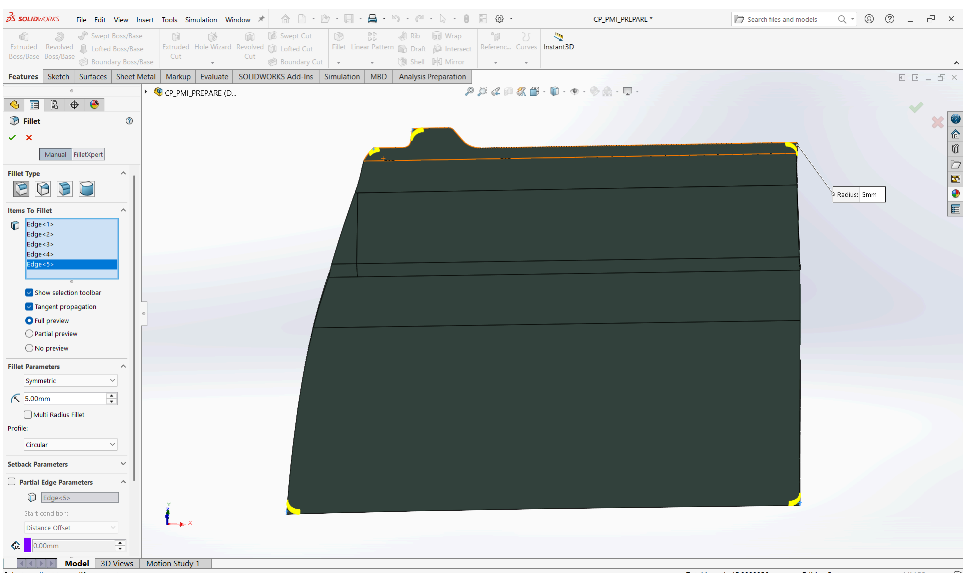


*Image 19: Adding Fillets to Protective Cover*

## Step 12: Add Any Features Required for Cover to Attach to Car Part

Might be necessary to add a hook or a push pin in order to keep the cover attached to the car while in use. This example would include a hook to attach to the rest of the door.

**Step 13: Save As .stl**

In SOLIDWORKS, locate the “File” tab in the top menu, click “Save As” (Dassault Systèmes, 2021). See *Image 20* and *Image 21* for reference.

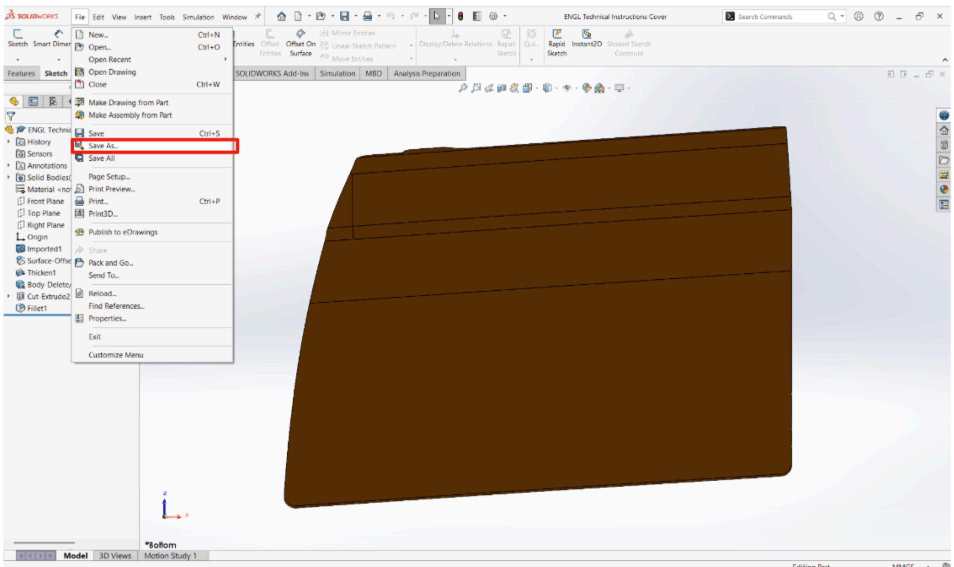

*Image 20: File → Save As*

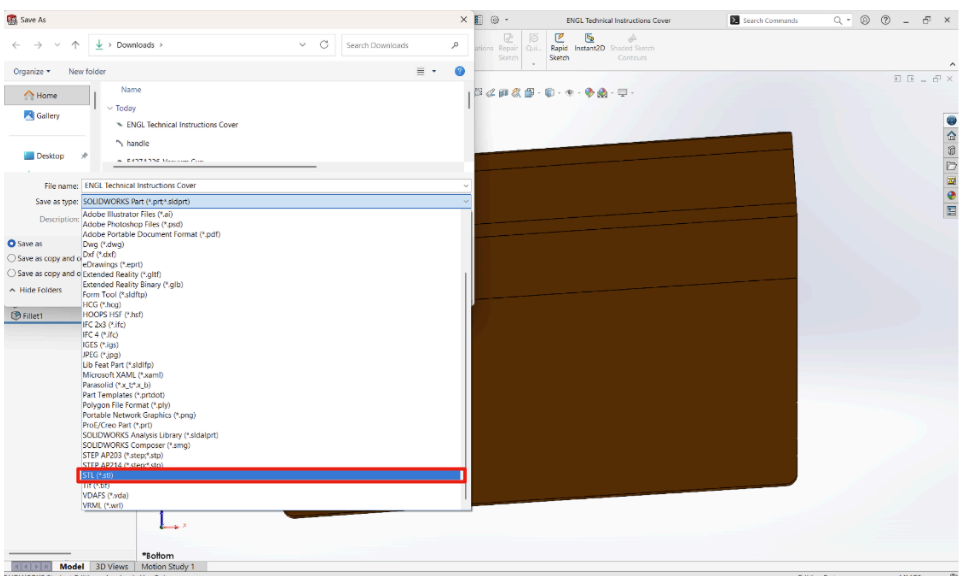

*Image 21: Save in .stl Format*

You now have a protective cover model that you can import into a slicer to upload to a 3D printer.

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